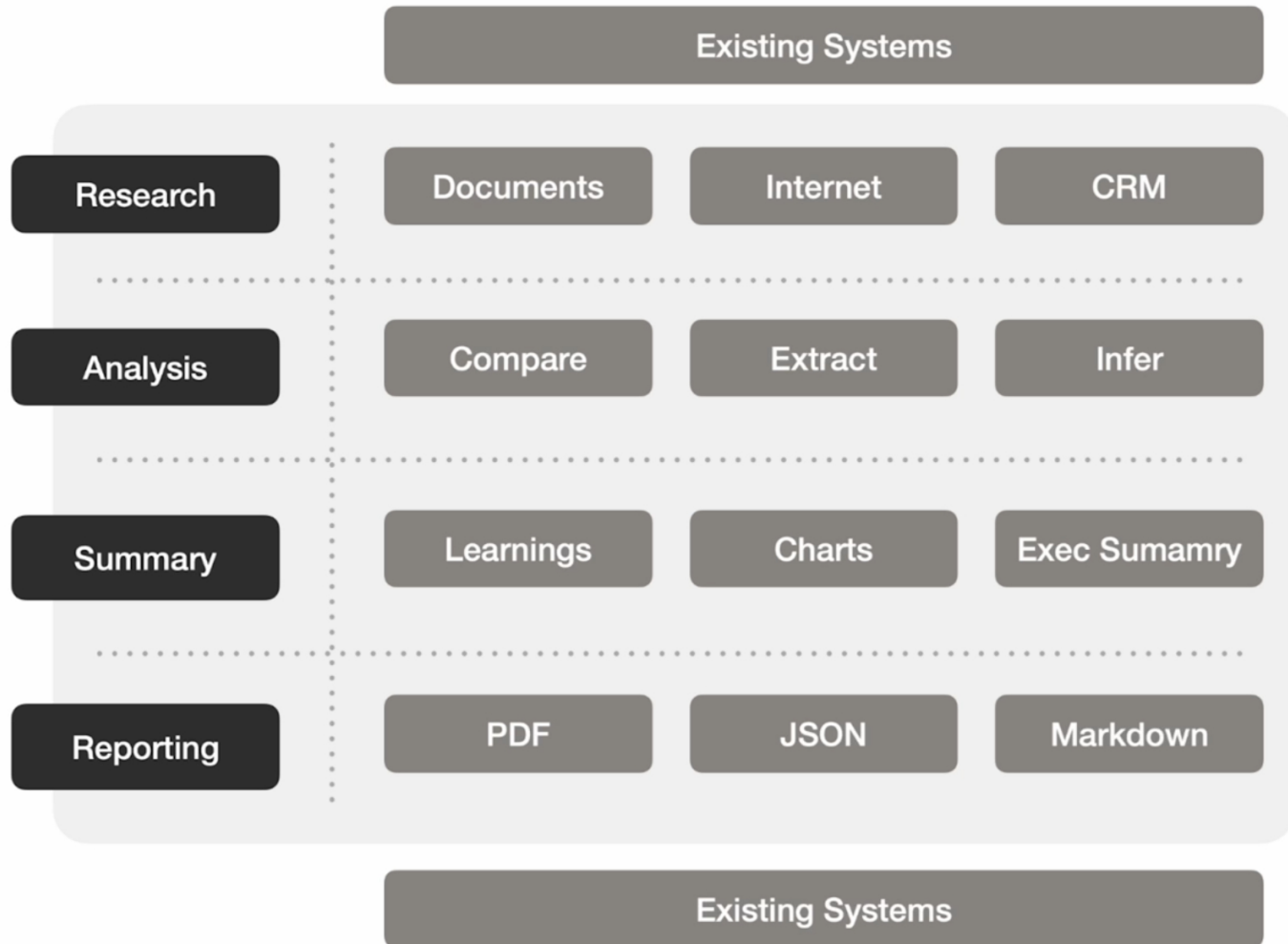


Practical Multi AI Agents and Advanced Use Cases with crewAI

Overview of Multi AI Agent Systems



DeepLearning.AI



Quick Recap:

What Are Multi AI Agent System and how to build them

Regular Apps

vs

AI Apps

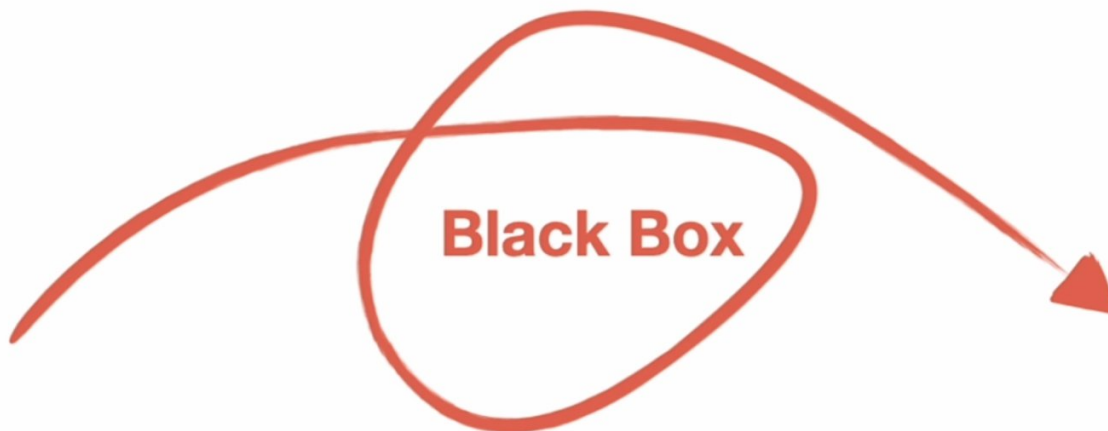
Well-defined
Transformations

Strong Typed
Inputs



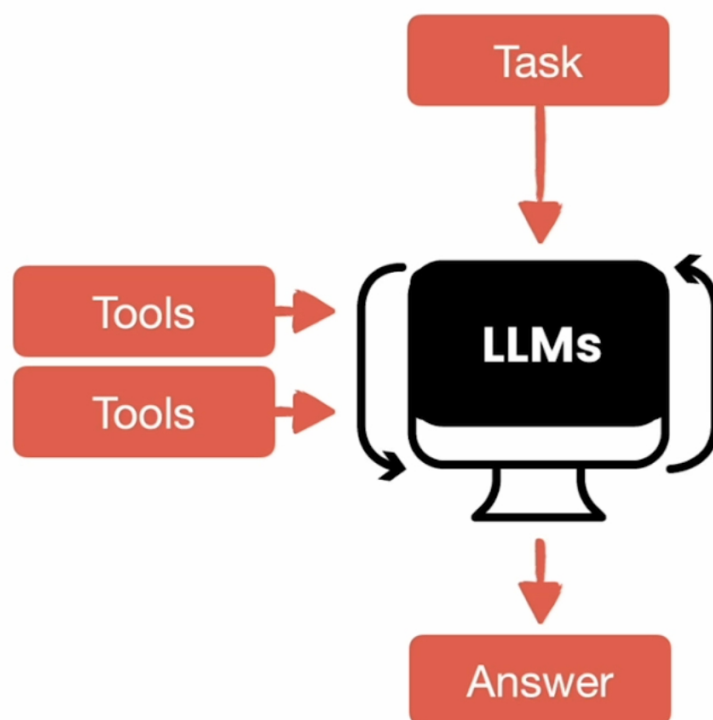
Strong Typed
Outputs

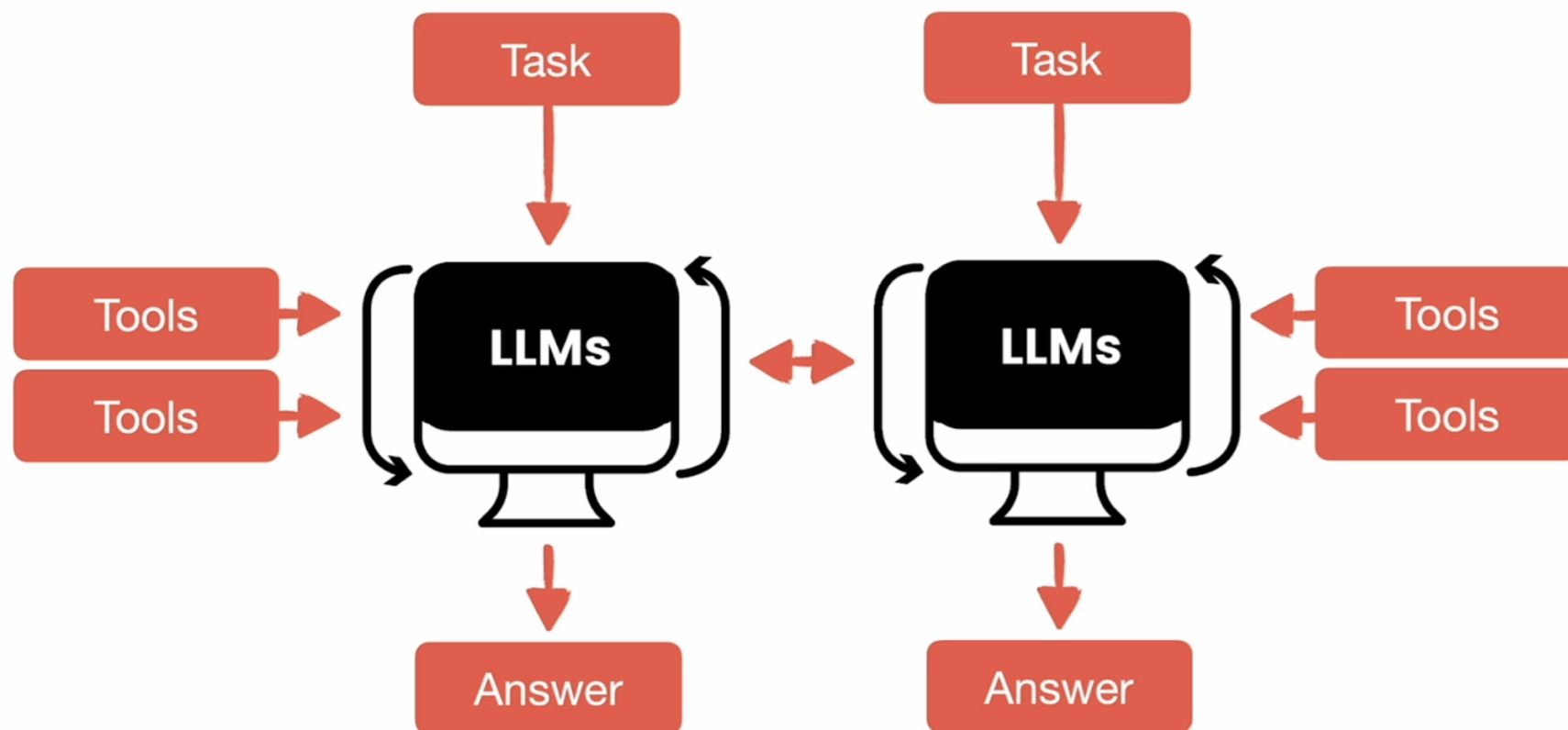
Fuzzy
Inputs

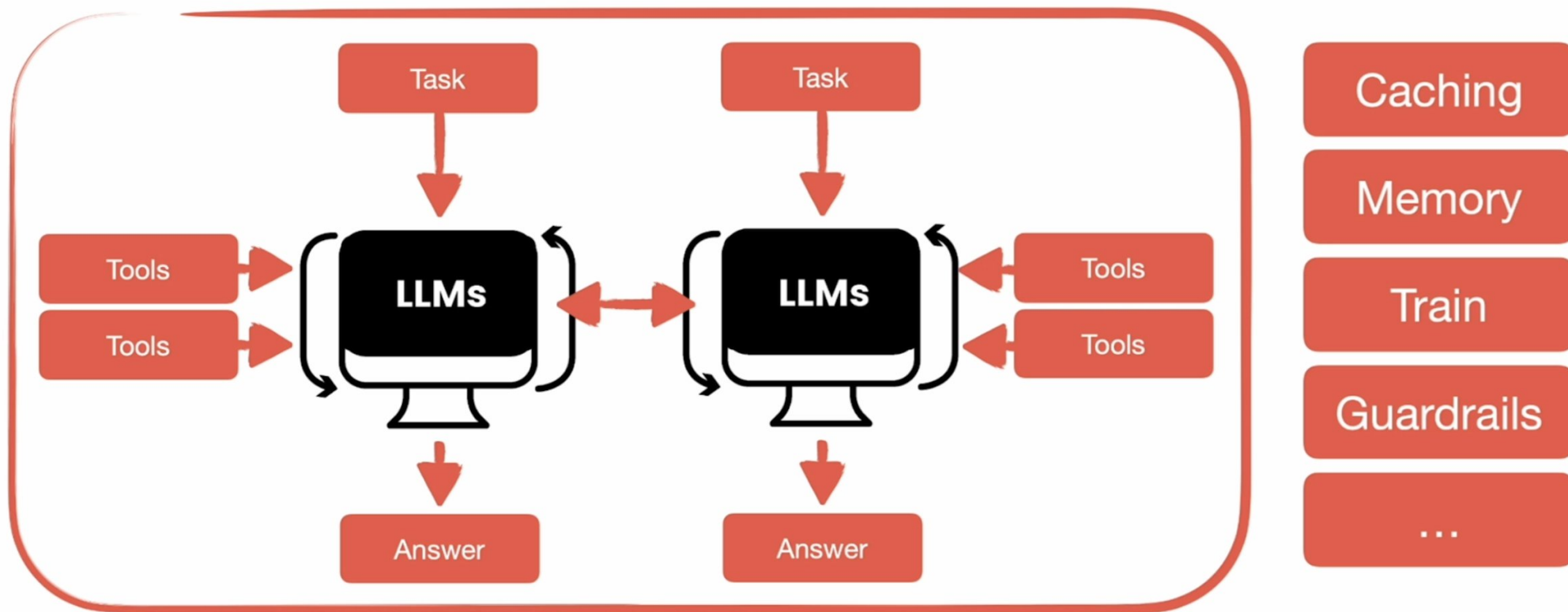


Fuzzy
Outputs

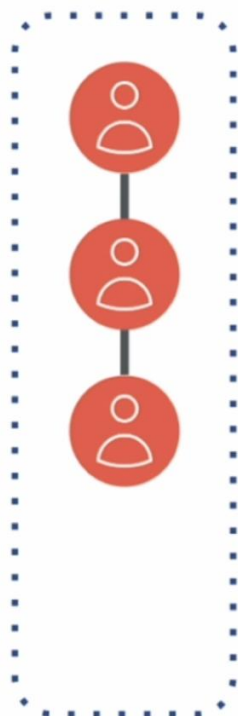
Multi-AI Agent Systems are a kind of AI app



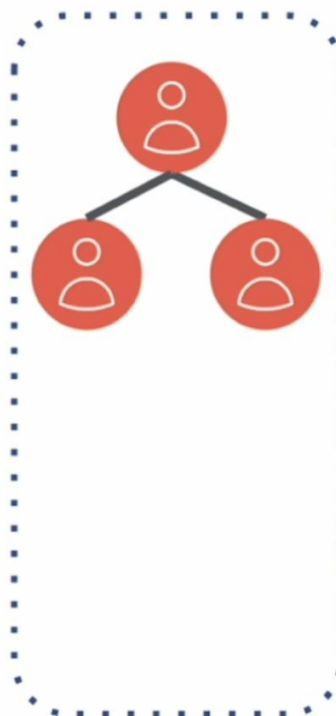




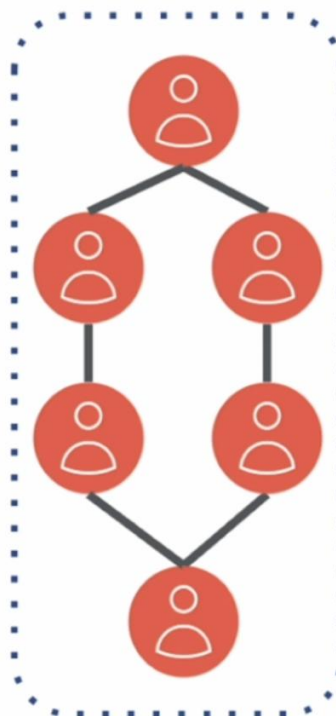
Sequential



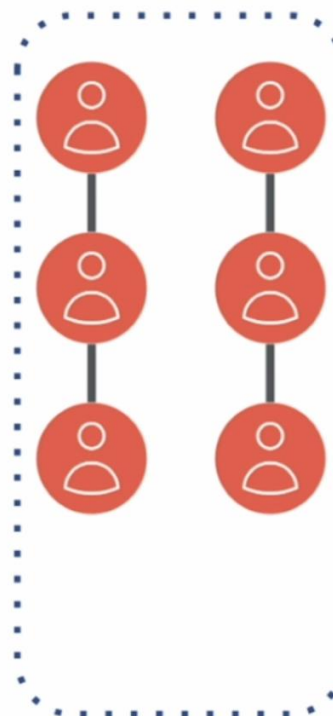
Hierarchical



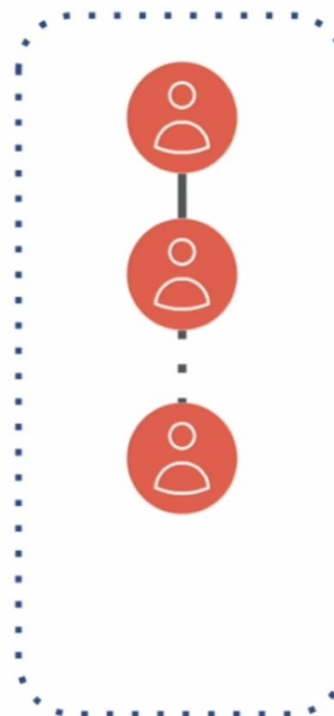
Hybrid

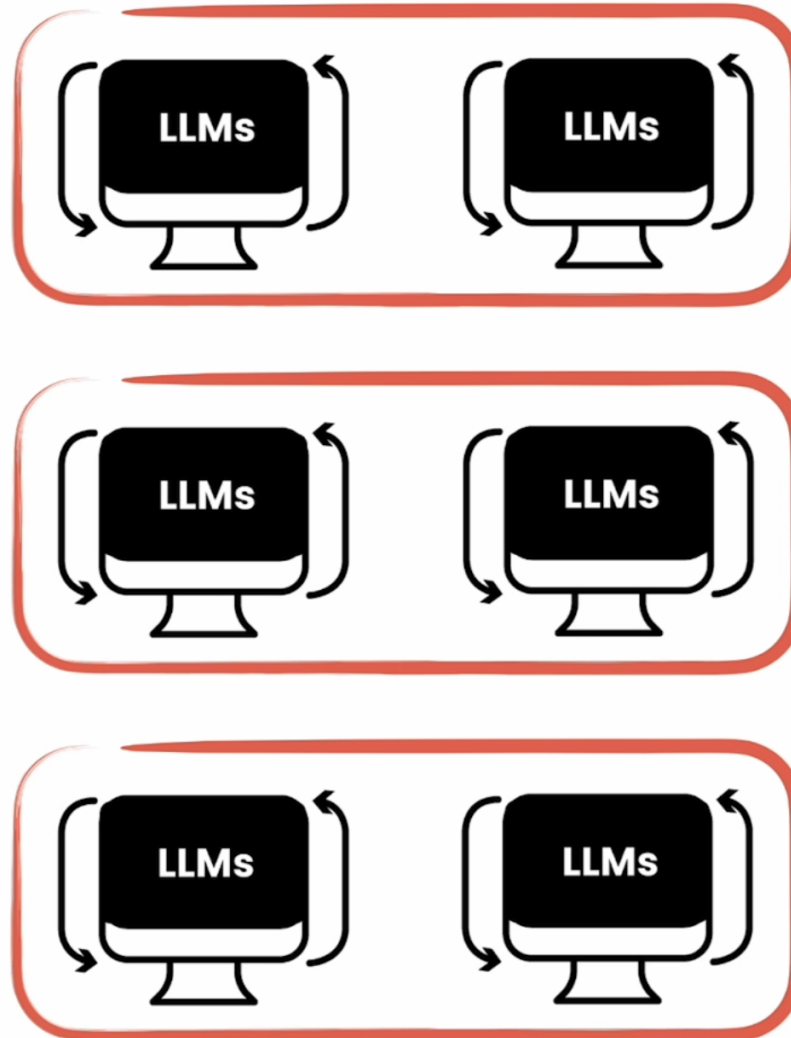


Parallel



Async

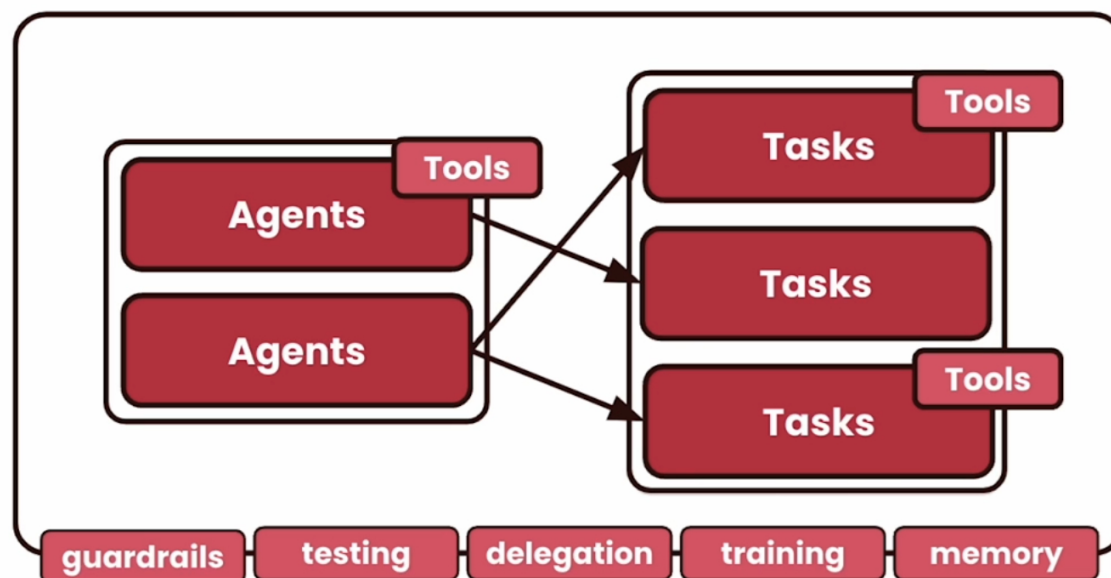




Main Building Blocks

for building Multi AI Agent Systems

Crew



Agents

- Role
- Goal
- Backstory

```
researcher:  
  role: >  
    {topic} Senior Data Researcher  
  goal: >  
    Uncover cutting-edge developments in {topic}  
  backstory: >  
    You're a seasoned researcher with a knack for uncovering...
```

Tasks

- Description
- Expected Output
- Agent

```
research_task:  
  description: >  
    Conduct a thorough research about {topic}...  
  expected_output: >  
    10 points of the most relevant information about {topic}...  
  agent: researcher
```

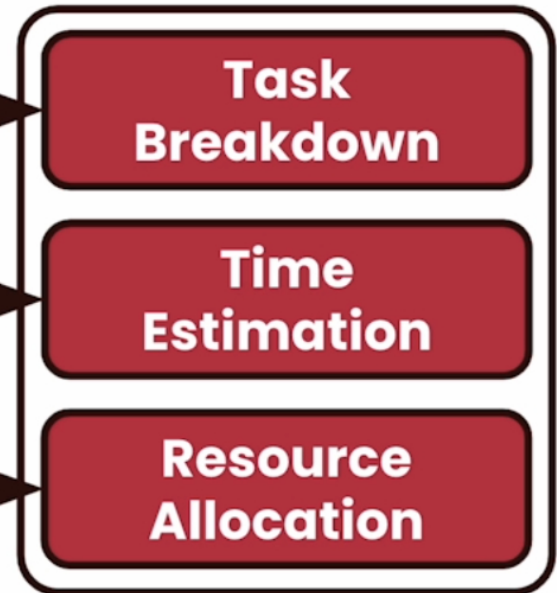
Lab 01

Automated Project: Planning, Estimation, and Allocation

Agents



Tasks



Project Plan

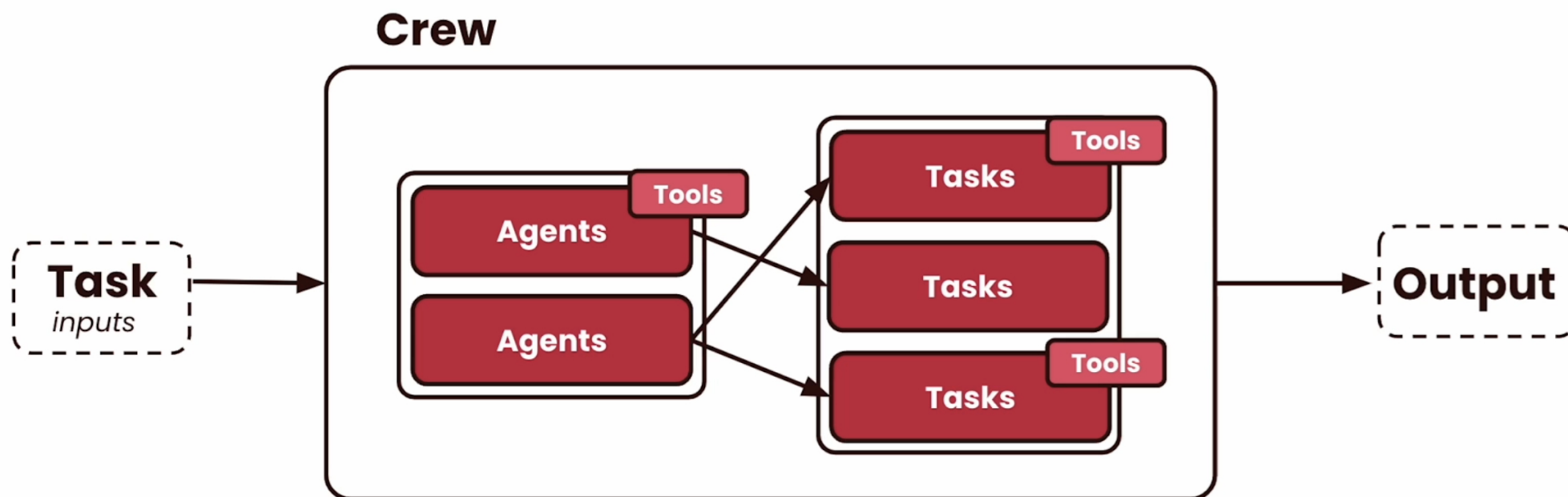
- Tasks / Allocations
 - Milestones
- 

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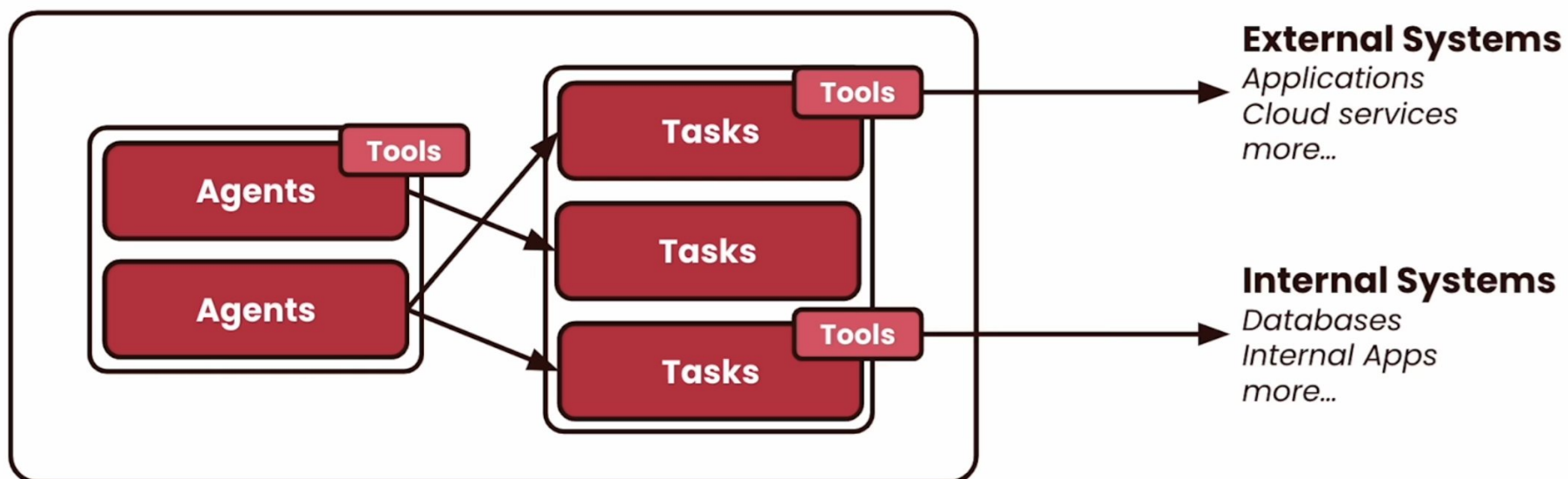
Internal and External Integrations



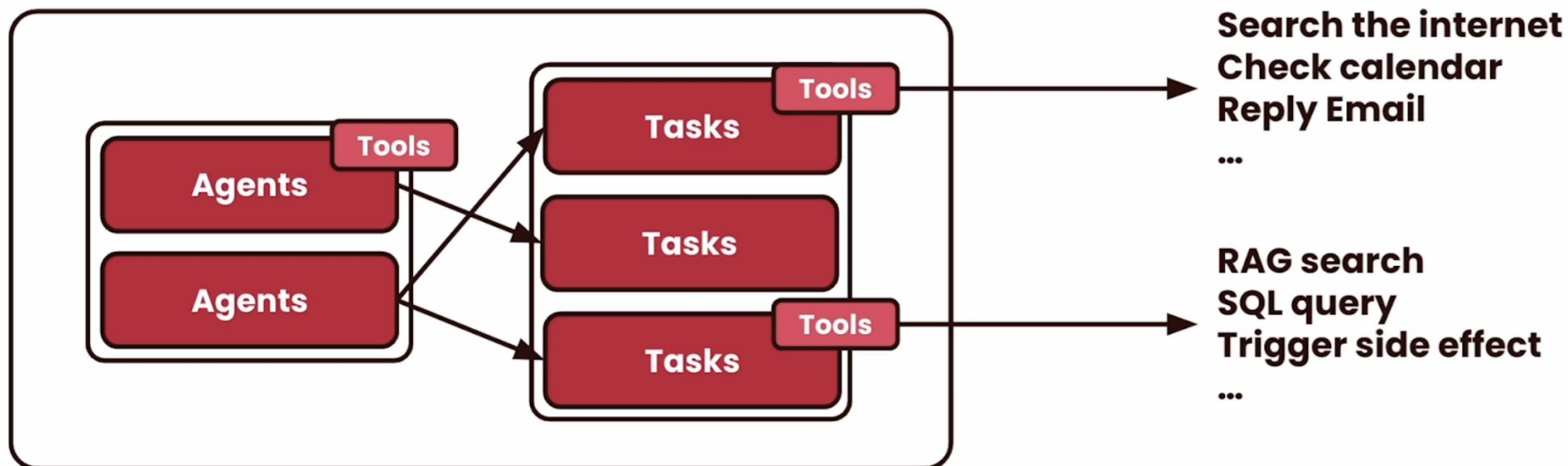
DeepLearning.AI



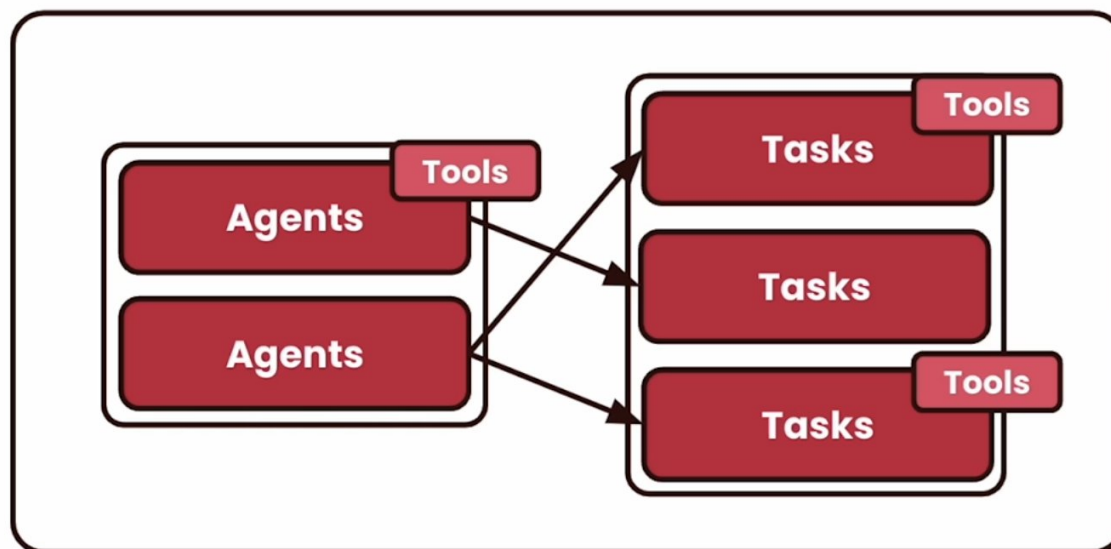
Crew



Crew



Crew



Write custom code

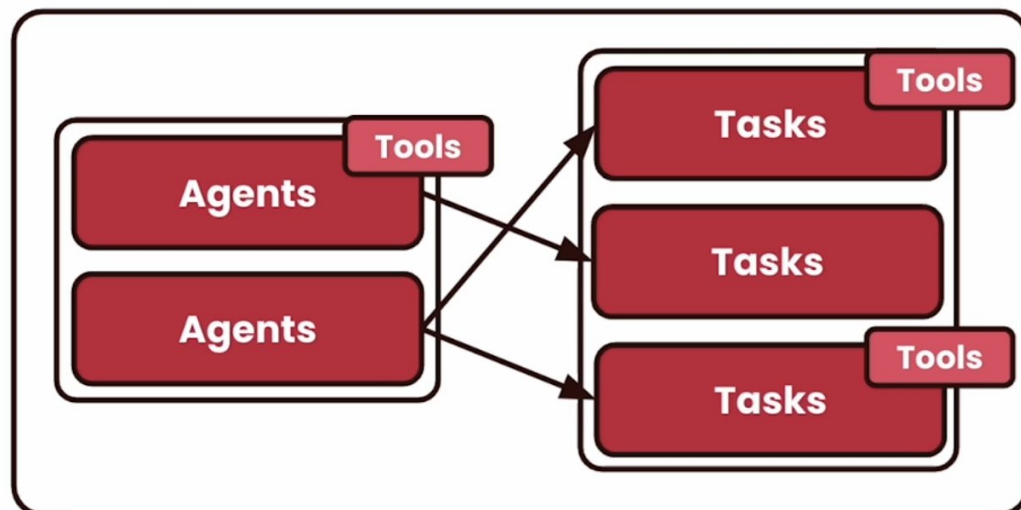
To try to accomplish what was not able with current tools

Real Time Reaction and self-healing capabilities

Example: Data Analysis Crew

Leveraging external and Internal data

Analysis Crew



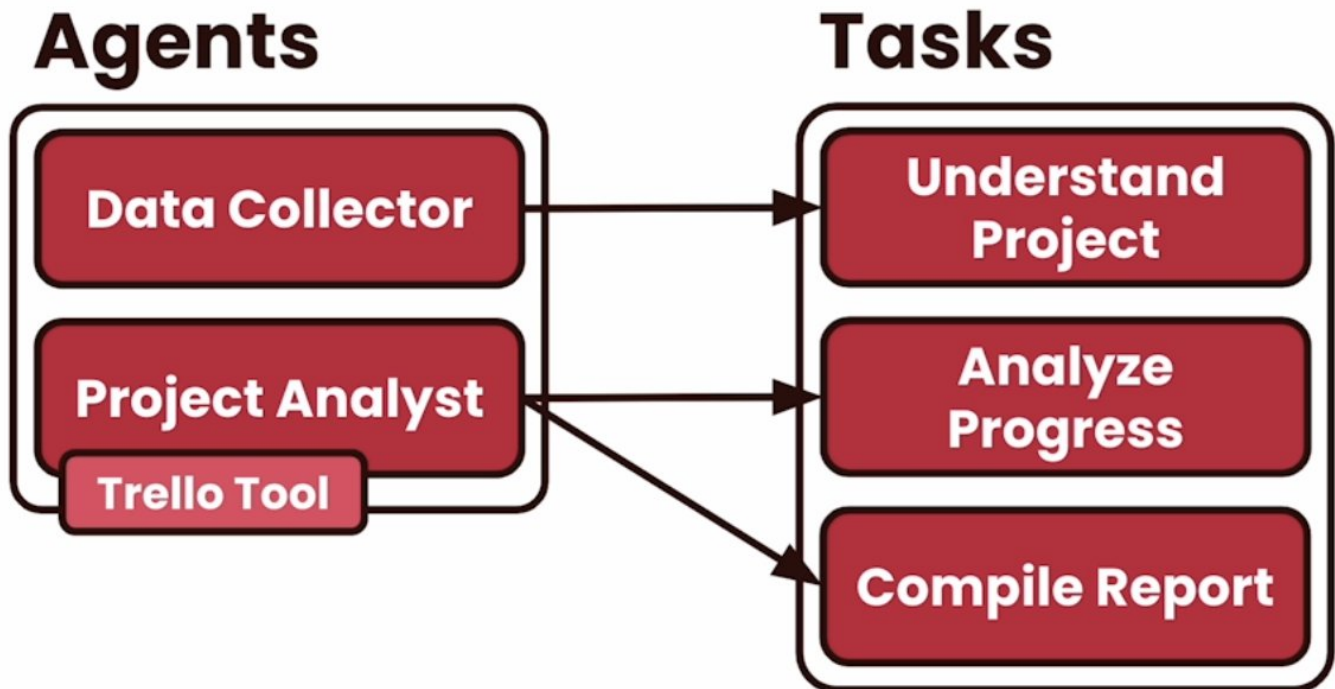
Crew Tasks

- External data sources for data on a company
- Cross with internal data of a company
- Put together full company records
- Fill missing data points with a RAG tool

Lab 02

External & Internal Integrations
Giving tools to your agents

Building progress report for project



Markdown Report

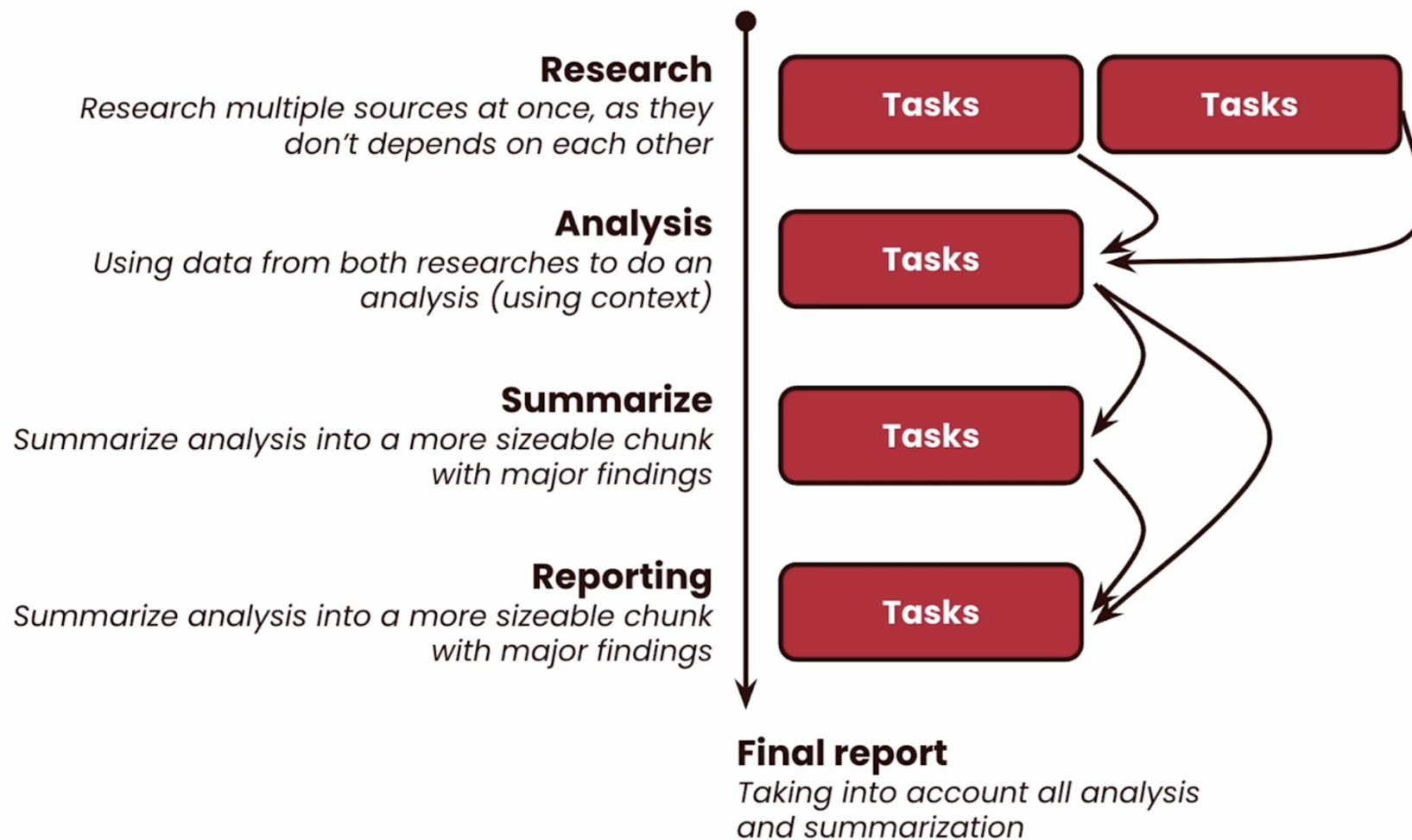
- Sprint Overview
- Task Summary
- Identified Issues and Blockers
- Progress and Delays
- Team Performance Review
- Action Items and Recommendations
- Anything else that is relevant to the project.

Practical Multi AI Agents and Advanced Use Cases with crewAI

Complex crew Setups



DeepLearning.AI

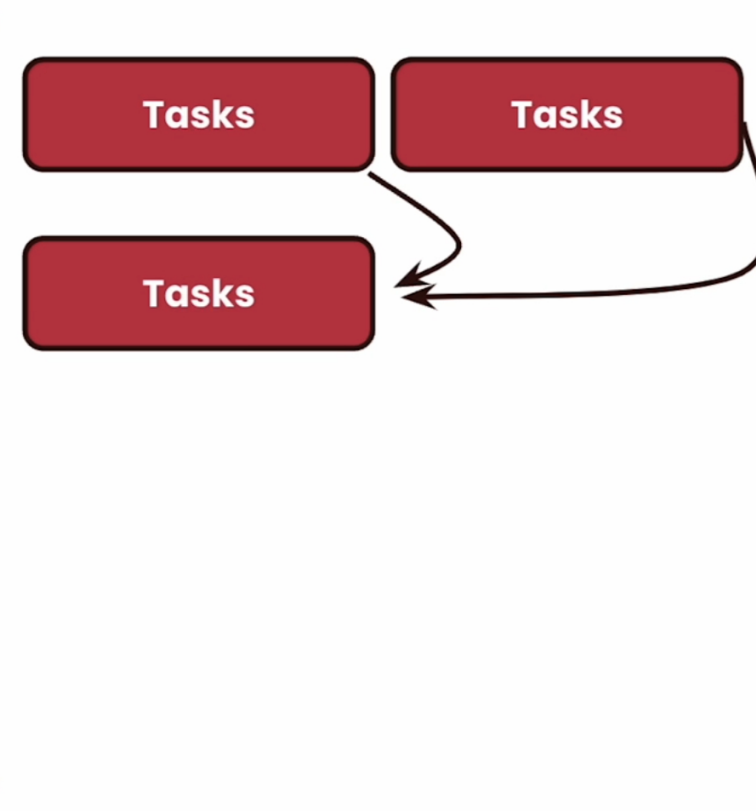


```
task = Task(  
    description='Research the project and...',  
    agent=analysis_agent,  
    async_execution=True,  
    expected_output="A summary of the..."  
)
```

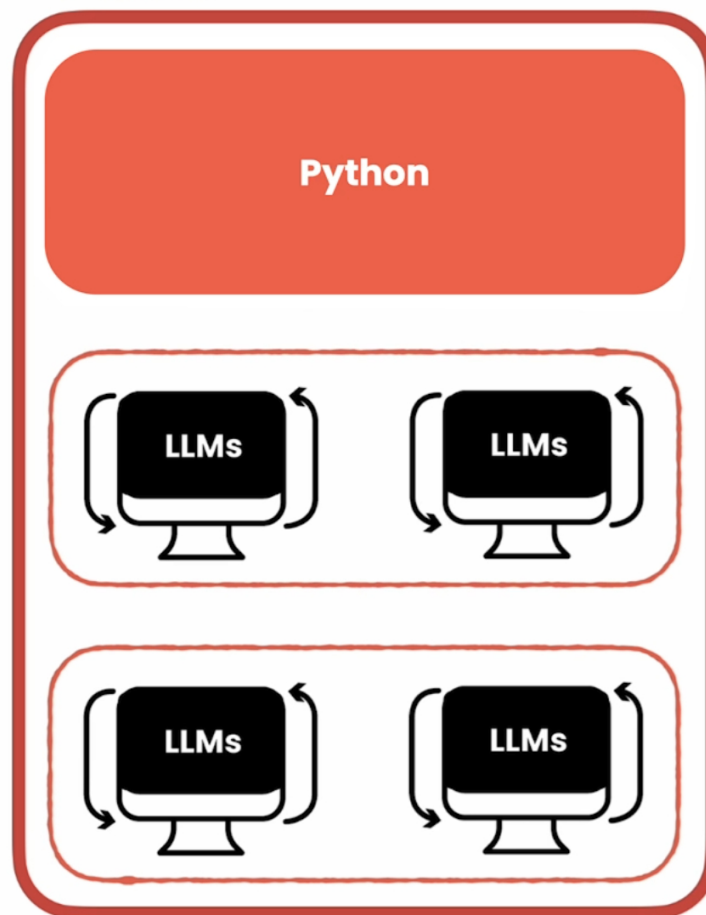
Tasks

Tasks

```
task = Task(  
    description='Analyze the...',  
    agent=analysis_agent,  
    expected_output="A complete analysis...",  
    context=[research_task_1, research_task_2]  
)
```



Flows

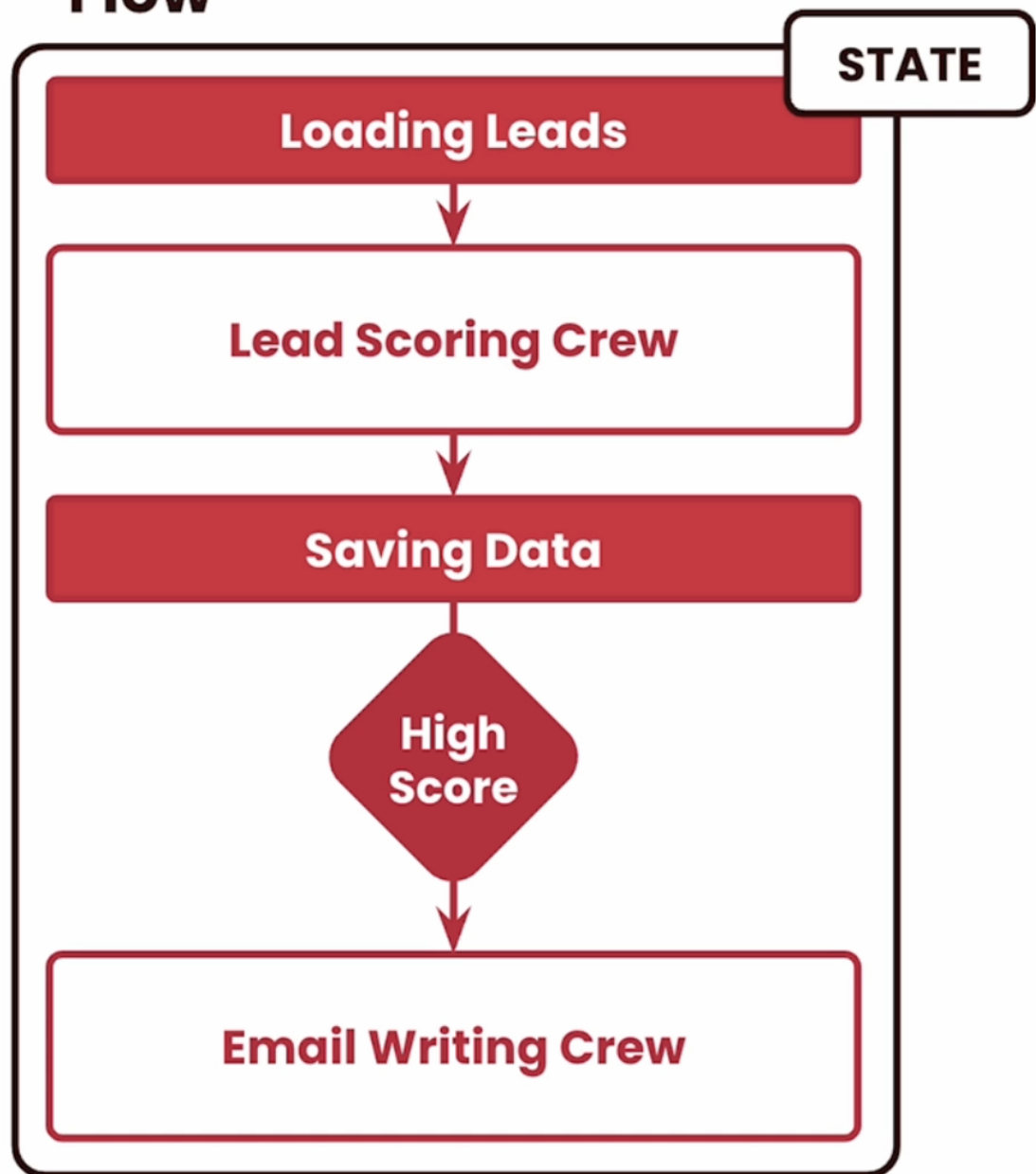


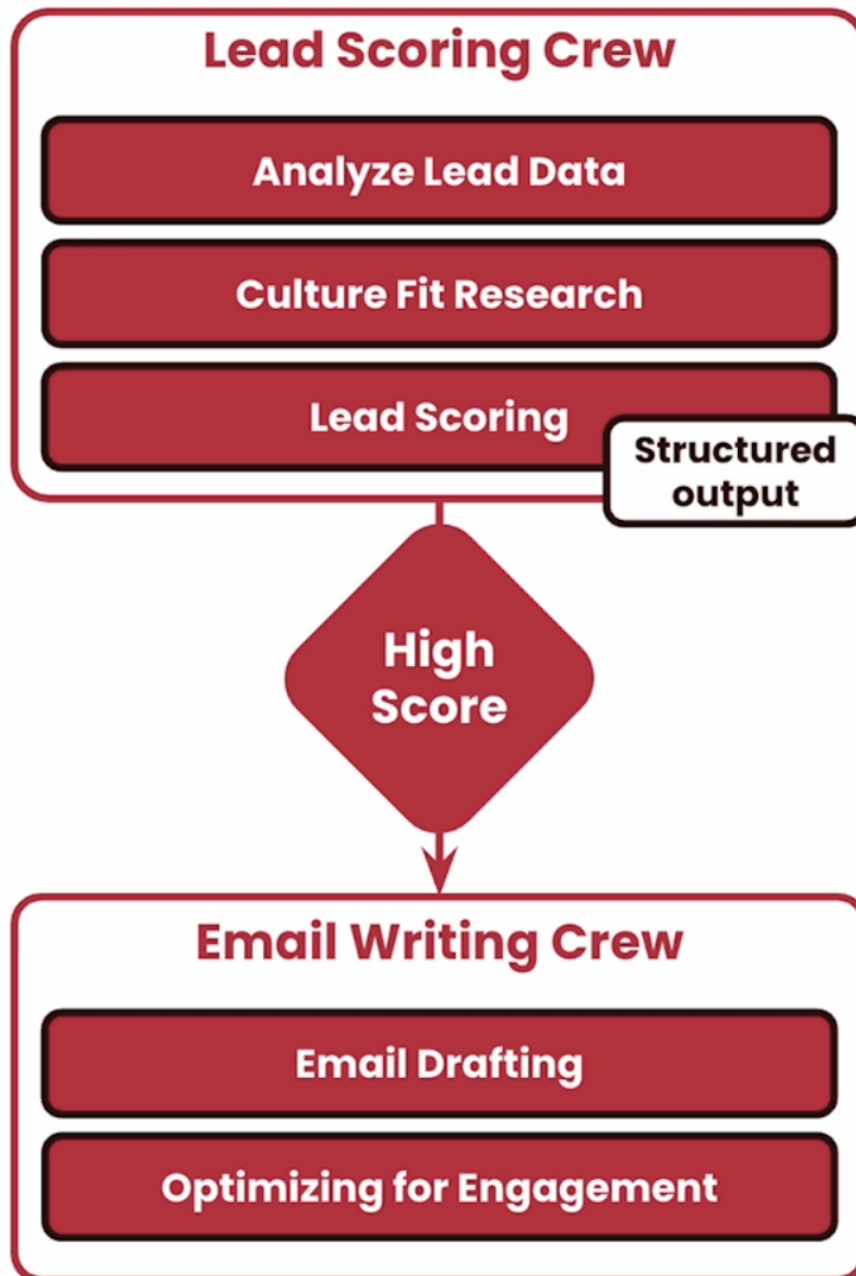
Lab 03

Complex Crew Setups Building Flows

Agentic Sales Pipeline

Flow





Sales Pipeline

- Research a potential lead
- Score them given the person and company
- If a qualified lead then a proper initial email

Practical Multi AI Agents and Advanced Use Cases with crewAI

Performance Optimization



DeepLearning.AI

Smaller models

Smaller models are cheaper to run as they require smaller servers and run faster

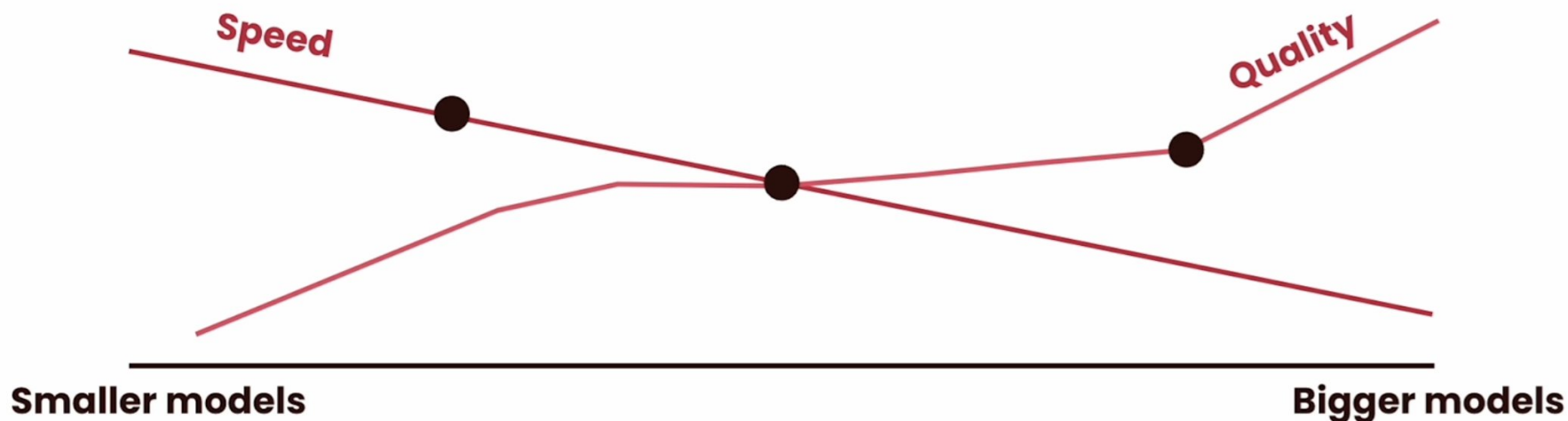
Speed

Quality

Bigger Models

Bigger models usually have more capable reasoning and can better adhere to formats

Consistency





Testing your Agent's Performance

Using crewAI Testing Features

Tasks

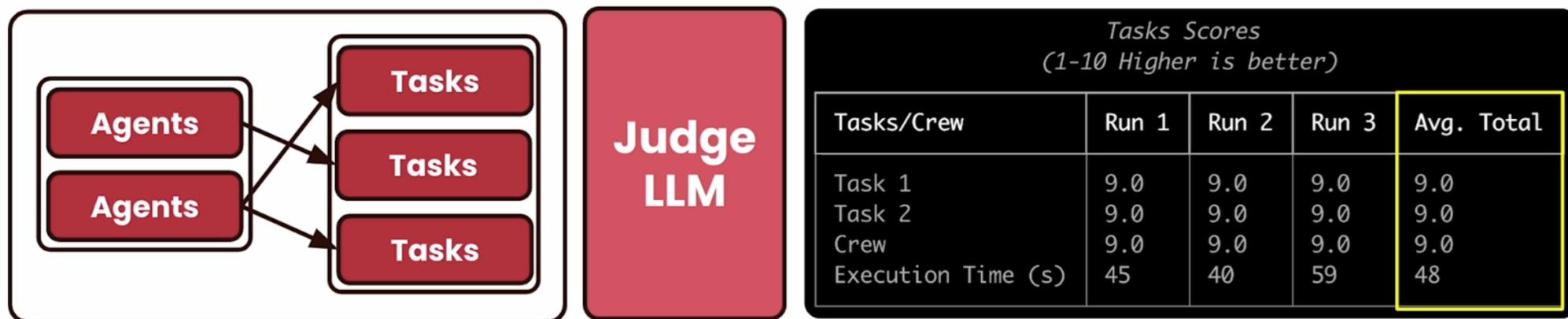
- **Description** → Compare the actual
- **Expected Output** → result with the expectation
- Agent

Tasks



```
$ crewai test
```

Crew



Tasks

- **Description** → Compare the actual result with
- **Expected Output** → the expectation
- Agent

Specific Format
Specific Style
Missing Info

Tasks

Tasks



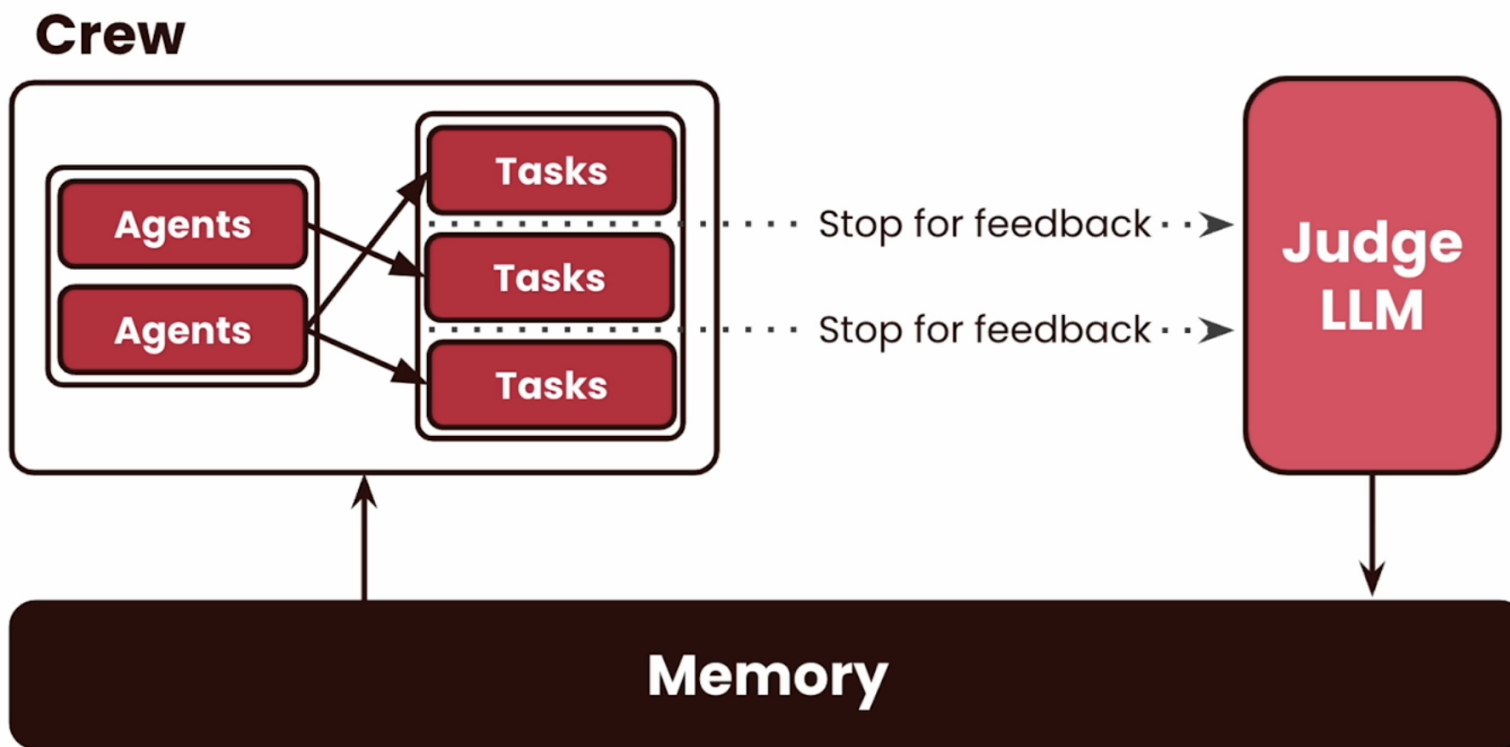


Training your Agent's Performance

Using crewAI Training Features



```
$ crewai train
```



Lab 04

Performance Optimization Support Data Insight Analysis

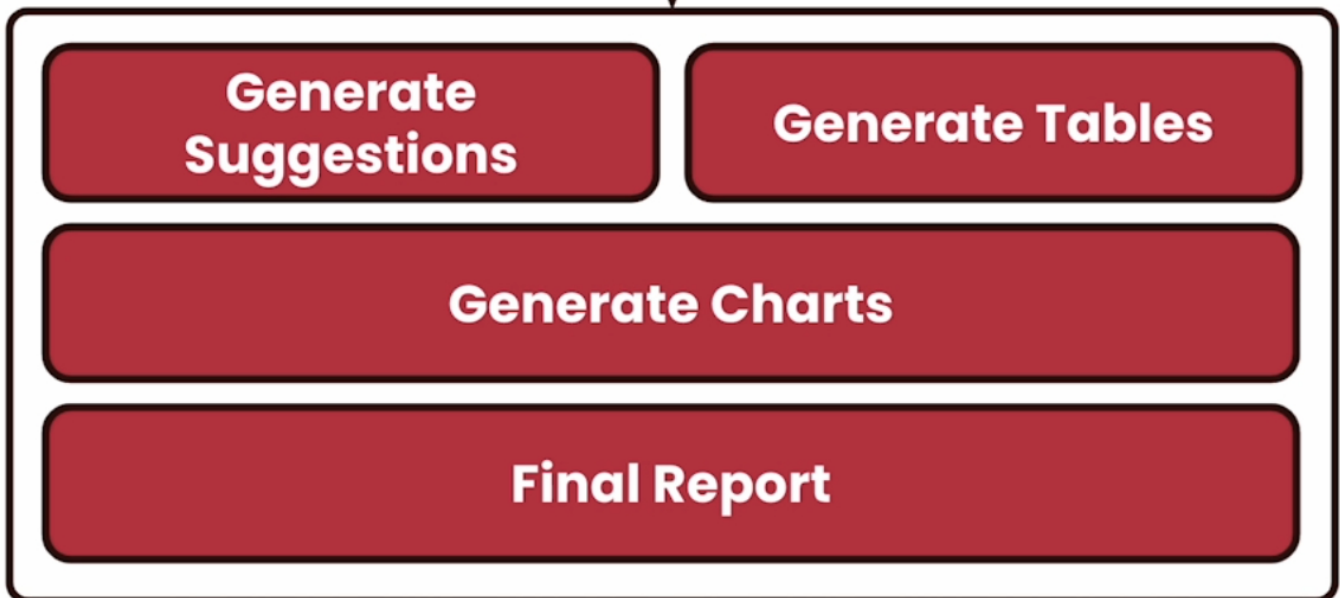
Support Data

- Who are the customers?
- Who is helping from our team?
- Issue types?
- Issue descriptions?
- Frustrations?

Agents



Tasks



Support Data Analysis

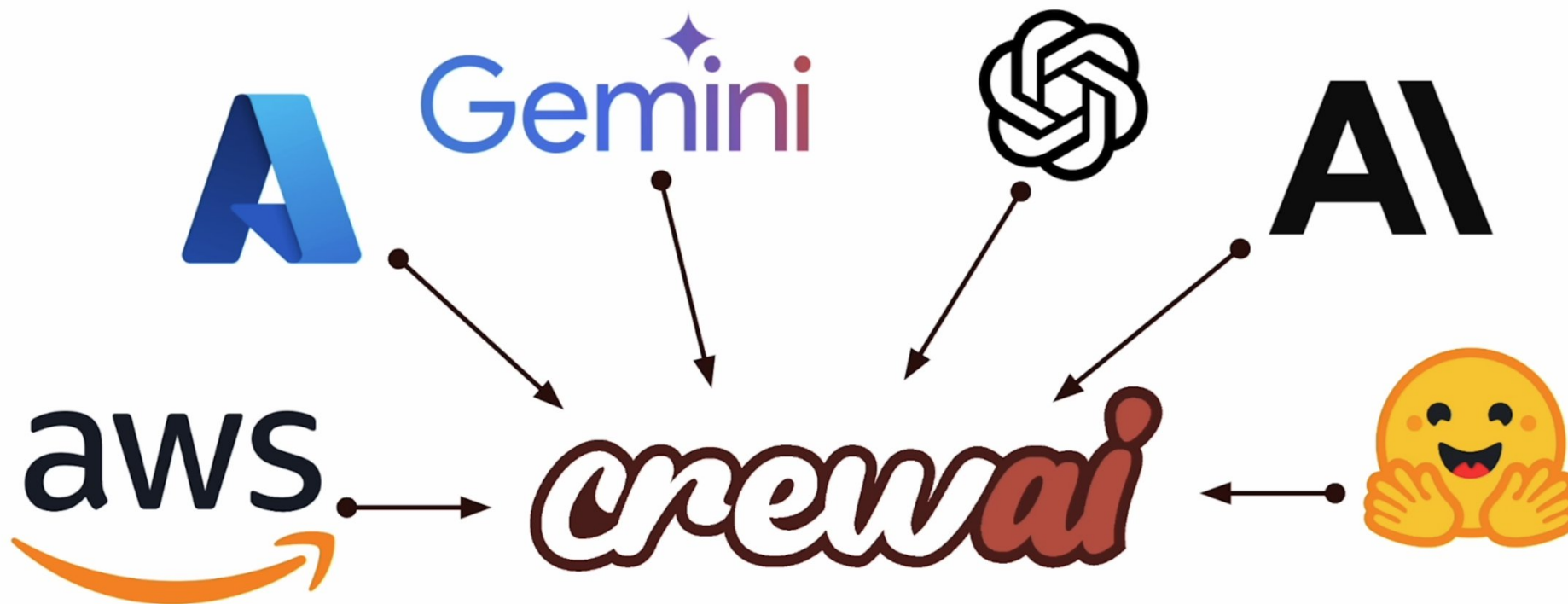
- Go over a series of data from support
- Generate suggestions for improvements
- Organize the data into tabular insights
- Plot charts to visualize trends
- Write a full final report on the analysis

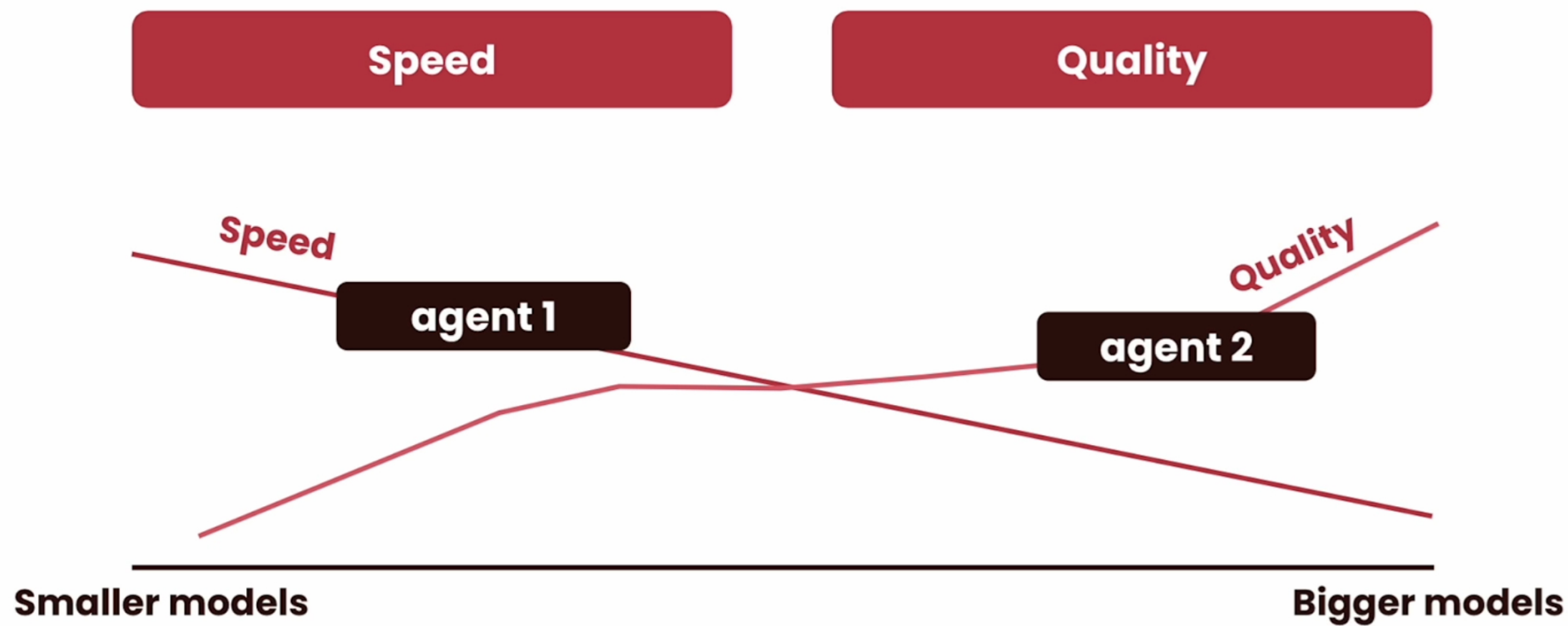
Practical Multi AI Agents and Advanced Use Cases with crewAI

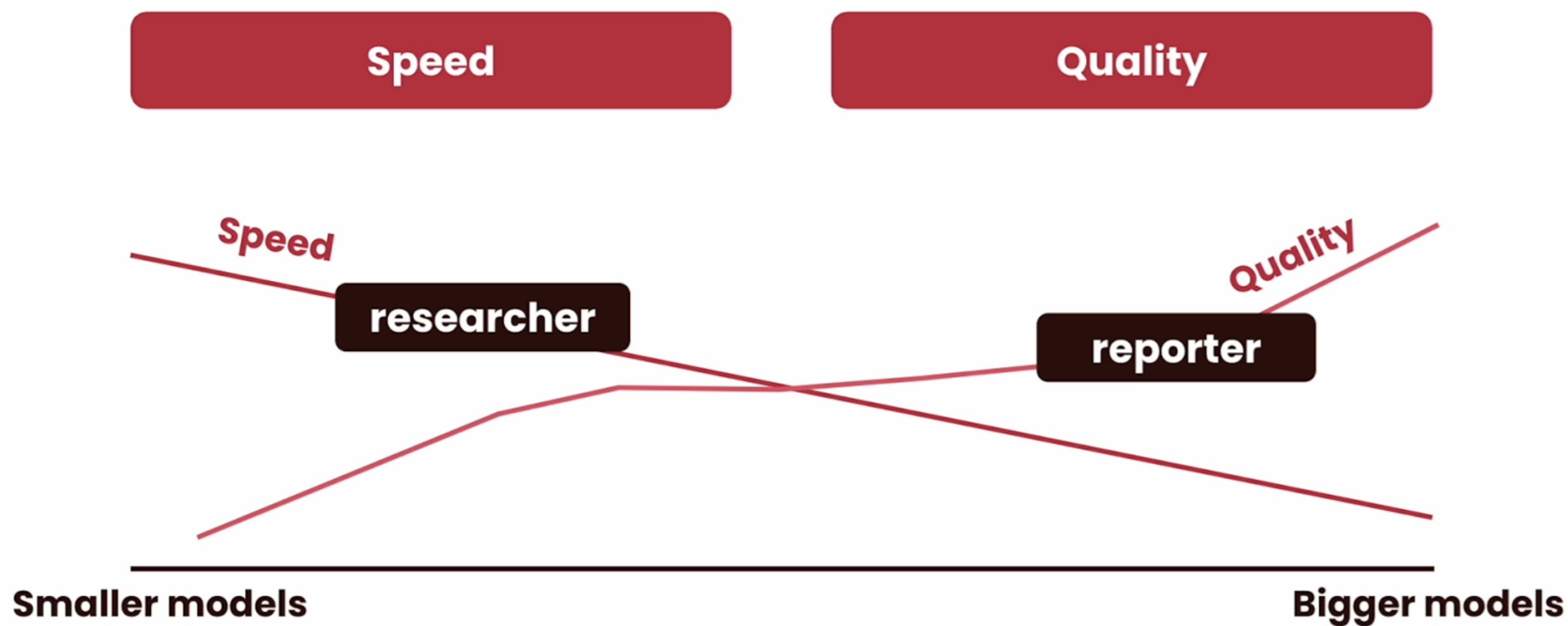
Multi-Model Use Cases

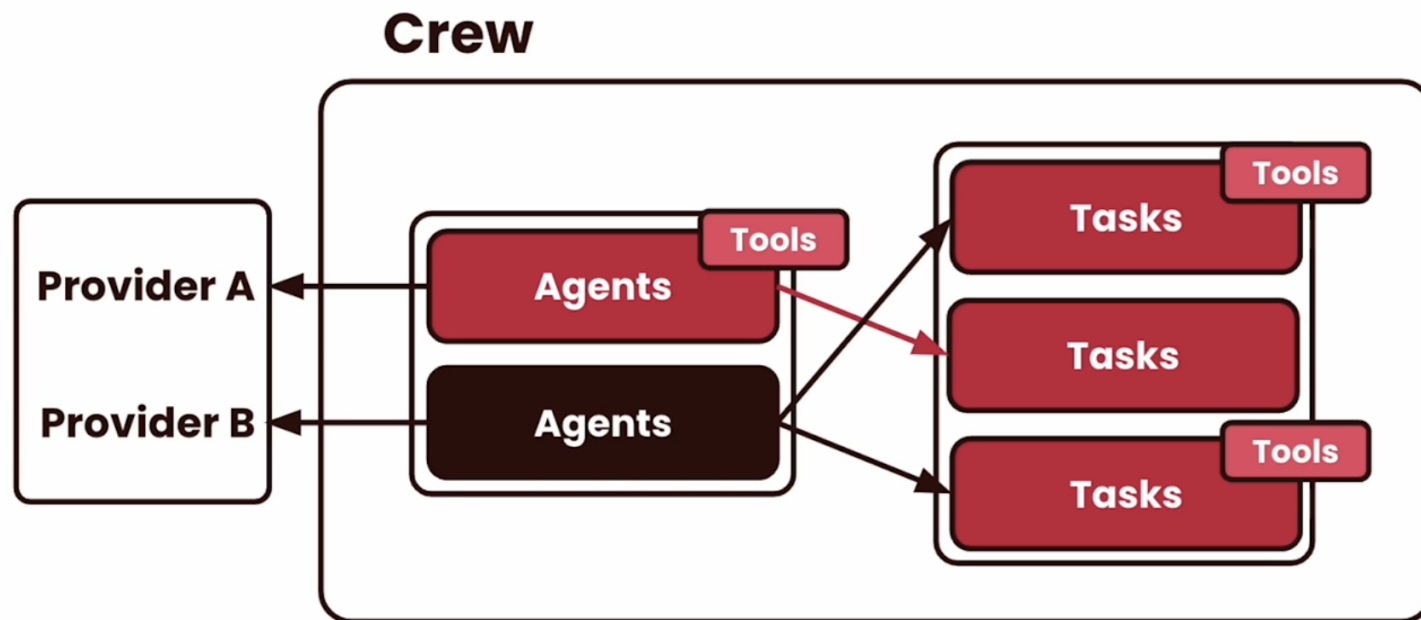


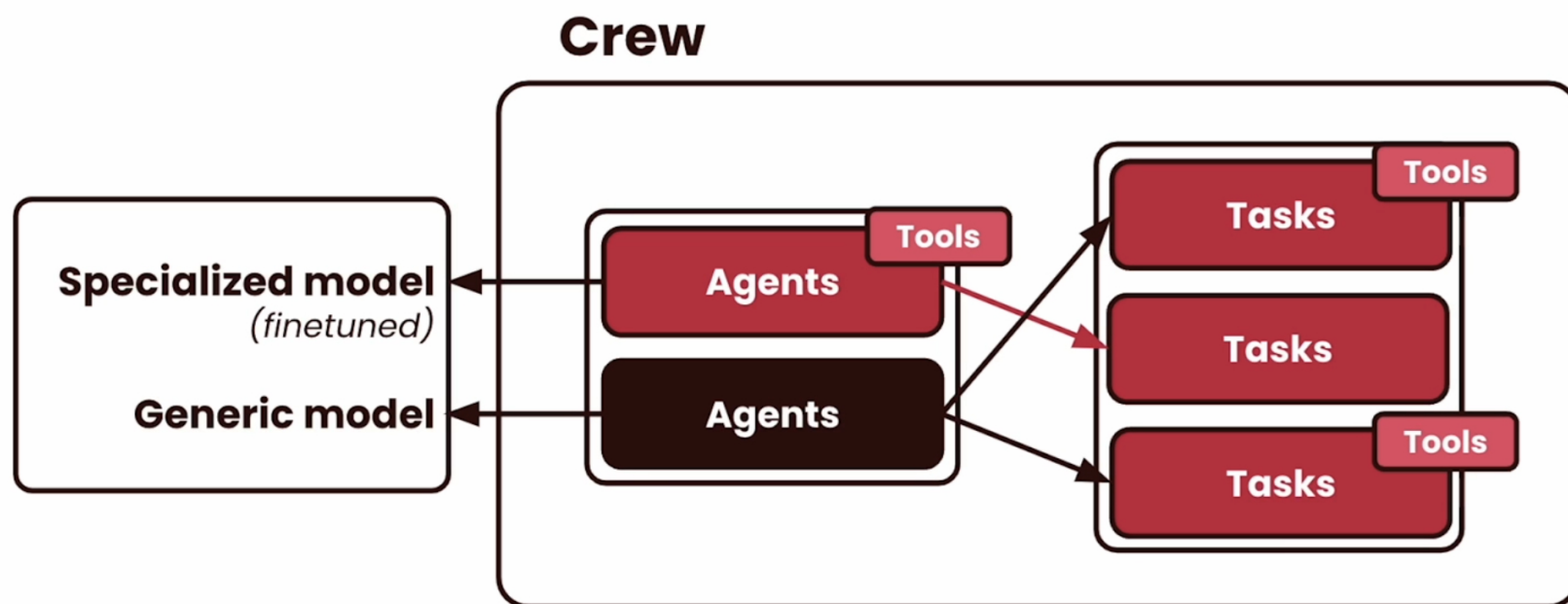
DeepLearning.AI





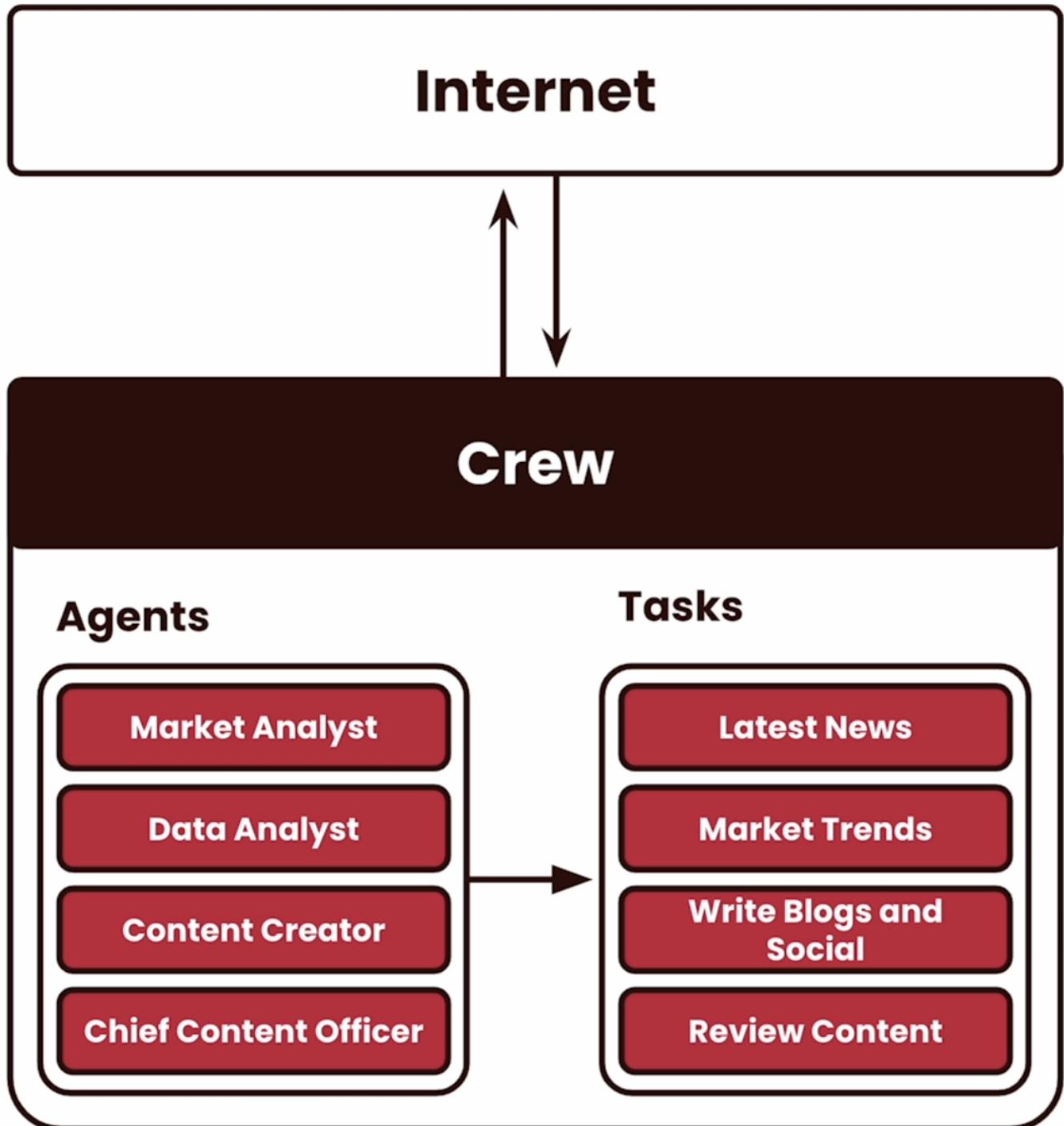


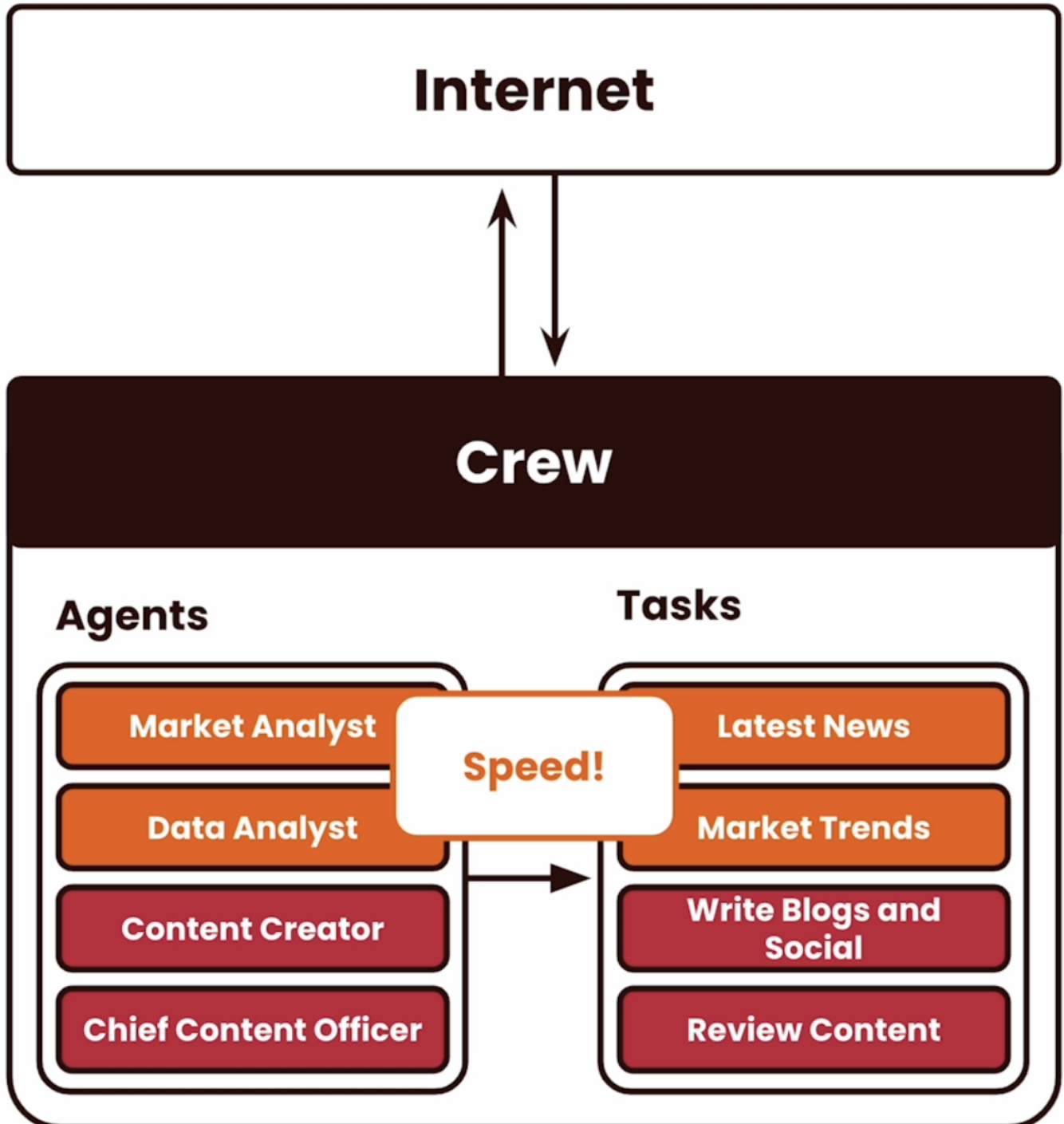


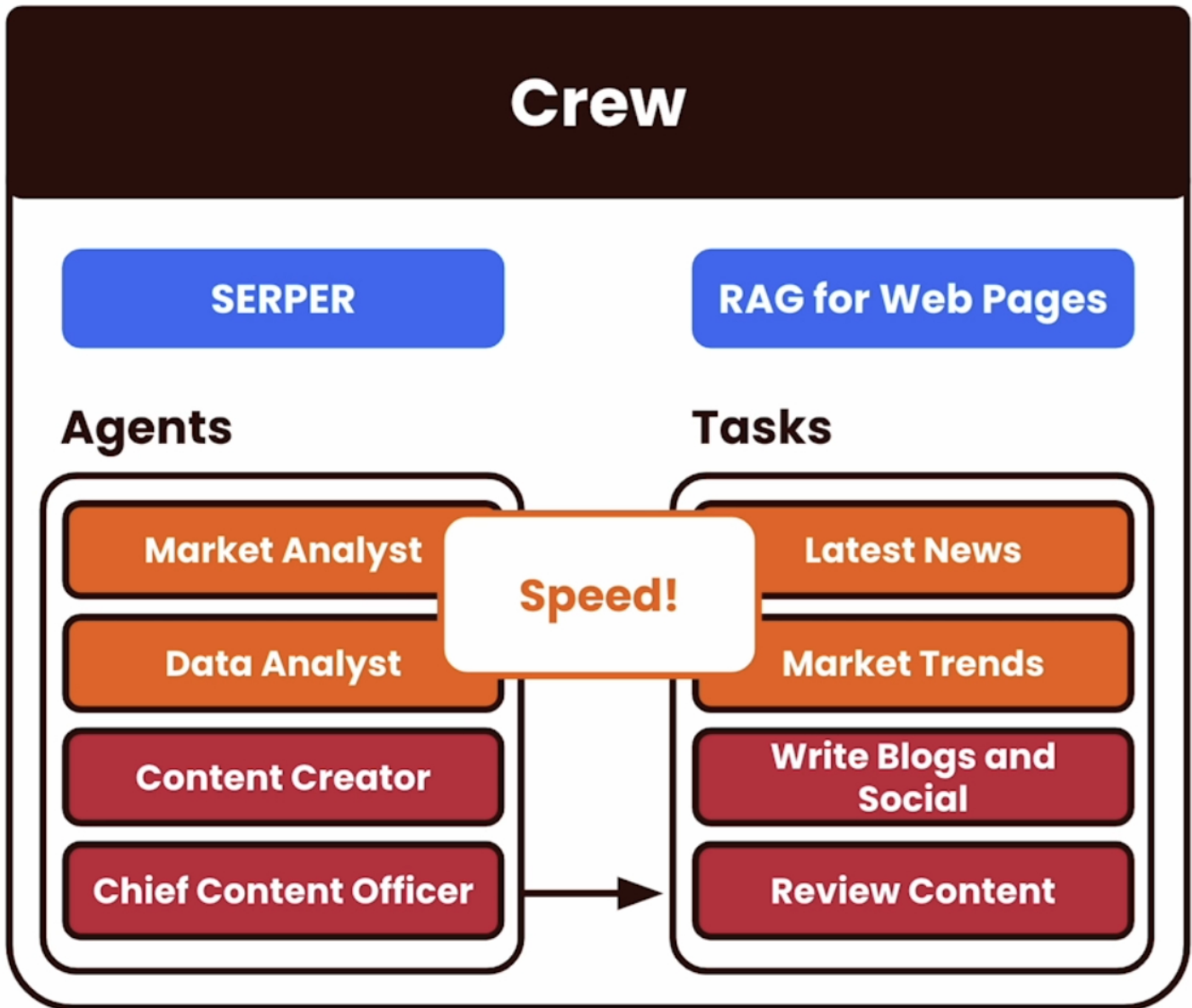


Lab 05

Multi-Model Use Cases Content Creation at Scale







Content Creation at Scale

- Search for the latest news around a financial topic
- Search and analyze any market data around it
- Create content for social media
- Create a full blog post on the topic
- Review the blog to make sure it's good

Practical Multi AI Agents and Advanced Use Cases with crewAI

Generate, Deploy and Integrate Crews



DeepLearning.AI

Starting a Crew from Scratch

How to get your environment started


```
$ crewai create crew <project>
```

```
new_project/
├── .gitignore
├── pyproject.toml
├── README.md
└── src/
    ├── new_project/
    │   ├── __init__.py
    │   ├── main.py
    │   ├── crew.py
    │   ├── tools/
    │   │   ├── custom_tool.py
    │   │   └── __init__.py
    │   └── config/
    │       ├── agents.yaml
    │       └── tasks.yaml
```

```
$ crewai install
```

Running Your Crews Locally

Best way to develop and iterate

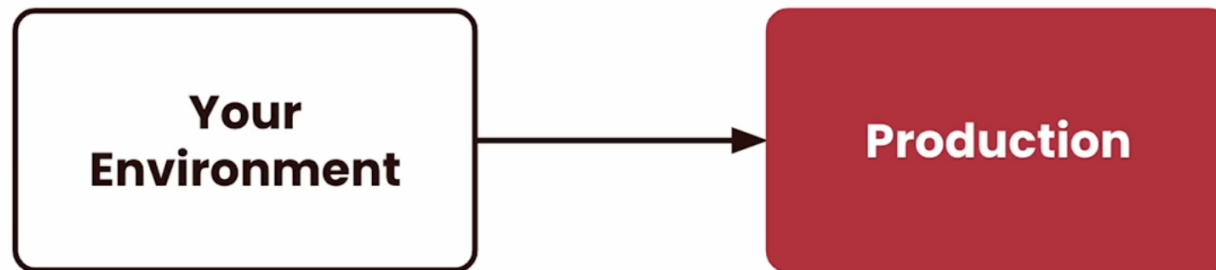
main.py

```
def run():  
    # ...  
  
def train():  
    # ...  
  
def replay():  
    # ...  
  
def test():  
    # ...
```

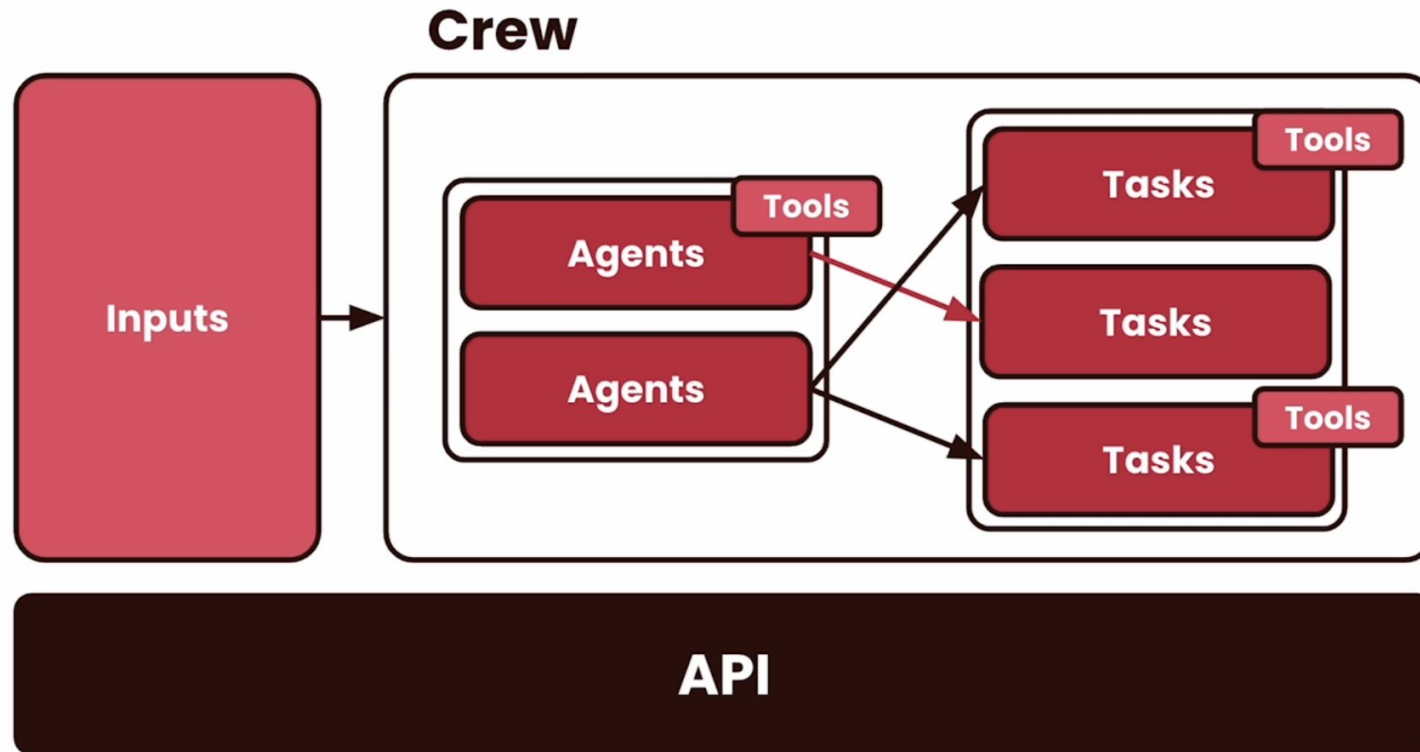
```
$ crewai run
```

Deploying your Crews

How to bring them to production



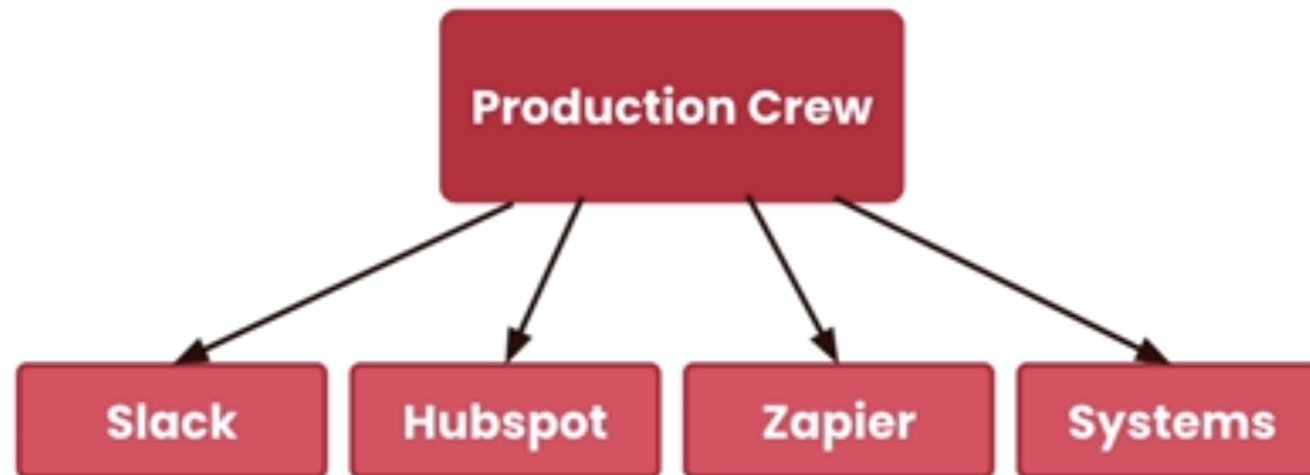

```
$ crewai deploy
```



<https://your-crew.crewai.com/kickoff>

Integrating Crews

Sharing with others!



<https://your-crew.crewai.com/kickoff>

```
{  
  "inputs": {  
    "topic": "AI LLMS"  
  },  
  "meta": "....",  
  "taskWebhookUrl": "https://...",  
  "stepWebhookUrl": "https://...",  
  "crewWebhookUrl": "https://..."  
}
```

[https://your-crew.crewai.com/**kickoff**](https://your-crew.crewai.com/kickoff)

[https://your-crew.crewai.com/**inputs**](https://your-crew.crewai.com/inputs)

[https://your-crew.crewai.com/**status/{run_id}**](https://your-crew.crewai.com/status/{run_id})

Lab 06

Generate, Deploy and Integrate Crews

Blog Post Crew in Production

Building our first Crew from the scratch!

- Will use the CLI to create a new crew
- Understand the file structure
- Install all required dependencies
- Execute our crew